Transfer: thermal energy

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Learning target: Choose an energy-transfer model and communicate its limits.

Accept equivalent correct methods and clearly labeled models. Language supports do not change the physics demand. Supplied data are examples; actual lab measurements and teacher checkpoints must be recorded from the lesson.

01. Warm-up: bring back an earlier idea *Review: Unit 5, Day 9.*
30% efficiency; 350 J rejected, assuming a complete cycle.

02. Worked example: explain the model
120 in=50 work out +70 stored.

03. Read and interpret the model
Flow from block to water; final common temperature lies between initial values under normal mixing assumptions; exact value needs heat capacities.

04. Apply the physics
800 J warming; 600 J phase change.

05. Explain your reasoning
350 J; efficiency 30%, must also satisfy second-law bound for reservoir temperatures.

06. Review: connect an earlier idea *Review: Unit 4, Day 22.*
6 W then 3 W. Same energy over twice the time gives half the power.

07. Exit check
+25 J.

08. STRETCH: test the model
Example work 20 J out requires heat 90 J in; infinitely many pairs share ΔU 70.

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01. Warm-up: bring back an earlier idea *Review: Unit 5, Day 9.*
Work 720 J; efficiency 30%.

02. Worked example: explain the model
120 in=50 work out +70 stored.

03. Read and interpret the model
Flow from block to water; final common temperature lies between initial values under normal mixing assumptions; exact value needs heat capacities.

04. Apply the physics
33,400+4200=37,600 J.

05. Explain your reasoning
350 J; efficiency 30%, must also satisfy second-law bound for reservoir temperatures.

06. Review: connect an earlier idea *Review: Unit 4, Day 22.*
v_{escape}=12 m/s. Assumptions include ideal central gravity, no atmosphere, no further thrust and negligible other bodies.

07. Exit check
Max efficiency 40%; max work 400 J; irreversibility/losses reduce actual work.

08. STRETCH: test the model
ΔU 400 J; Q 460 J. No phase change, suitable constant heat capacity and specified system accounting.